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United States Patent Application Publication

20250282010

Kind Code

A1

Publication Date

September 11, 2025

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TIGHTENING APPARATUS AND BATTERY MANUFACTURING SYSTEM

Abstract

The present application relates to a tightening apparatus and a battery manufacturing system. The tightening apparatus includes: a loading station, configured to bear an assembly to be tightened; a moving device, capable of moving relative to the loading station; a tightening device, arranged on the moving device; and a vision device, arranged on the moving device and configured to identify positions to be tightened of the assembly to be tightened on the loading station; where the moving device is configured to control the tightening device to move right above the positions to be tightened according to the positions to be tightened of the assembly to be tightened that are identified by the vision device, and the tightening device is configured to tighten tightening components to the positions to be tightened.

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Appl. No.: 19/211410

Filed: May 19, 2025

Foreign Application Priority Data

CN

202420292090.5

Feb. 18, 2024

Related U.S. Application Data

parent WO continuation PCT/CN2024/108891 20240731 PENDING child US 19211410

Publication Classification

Int. Cl.: B23P19/06 (20060101); H01M10/04 (20060101)

U.S. Cl.:

CPC B23P19/06 (20130101); H01M10/0404 (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present application is a continuation of International Application No. PCT/CN2024/108891, filed on Jul. 31, 2024, which claims priority to Chinese Patent Application No. 202420292090.5 filed on Feb. 18, 2024 and entitled “TIGHTENING APPARATUS AND BATTERY MANUFACTURING SYSTEM”, each are hereby incorporated by reference in their entirety.

TECHNICAL FIELD

[0002] The present application relates to the technical field of battery manufacturing, and in particular, to a tightening apparatus and a battery manufacturing system.

BACKGROUND

[0003] The statements herein only provide background information related to the present application and do not necessarily constitute the prior art.

[0004] A battery includes a case and a plurality of battery cells, and the plurality of battery cells are accommodated in the case. The case typically includes two parts: a case body and a cover. In the battery manufacturing process, the plurality of battery cells need to be accommodated in the case body first, then the cover is lidded with an opening of the case body, and the cover is connected to the case body with screws. In the related art, after the cover is positioned to the case body, a tightening gun is required to tighten the screws into the cover and the case body. However, the tightening failure rate of the screws is high and the tightening efficiency is low.

SUMMARY

[0005] In view of the issues, the present application provides a tightening apparatus and a battery manufacturing system, which can relieve the problems of high tightening failure rate and low tightening efficiency.

[0006] In a first aspect, the present application provides a tightening apparatus, which includes:

[0007] a loading station, configured to bear an assembly to be tightened; [0008] a moving device, capable of moving relative to the loading station; [0009] a tightening device, arranged on the moving device; and [0010] a vision device, arranged on the moving device and configured to identify positions to be tightened of the assembly to be tightened on the loading station; [0011] where the moving device is configured to control the tightening device to move right above the positions to be tightened according to the positions to be tightened of the assembly to be tightened that are identified by the vision device, and the tightening device is configured to tighten tightening components to the positions to be tightened.

[0012] The aforementioned tightening apparatus operates as follows: after the loading station bears the assembly to be tightened, the moving device may drive the vision device proximal to the assembly to be tightened to identify the positions to be tightened of the assembly to be tightened: according to the positions to be tightened, the moving device may control the tightening device to move to the positions to be tightened, and subsequently; the tightening device tightens the tightening components to the positions to be tightened of the assembly to be tightened, thereby completing the tightening operation. Since the positions to be tightened of the assembly to be

tightened are identified by the vision device before a tightening operation, the moving device is capable of accurately driving the tightening device to be right above the positions to be tightened to tighten the tightening components to the positions to be tightened of the assembly to be tightened with success. Therefore, the tightening failure rate of the tightening components is reduced, and due to the use of the tightening device that automatically tightens the assembly to be tightened, the tightening efficiency is also improved.

[0013] In some embodiments, the tightening device includes a plurality of tightening mechanisms, all of the tightening mechanisms are spaced apart from each other, and each tightening mechanism is capable of tightening the tightening component to the corresponding position to be tightened.

[0014] By the arrangement of the plurality of tightening mechanisms, a plurality of positions to be tightened of the assembly to be tightened can be tightened simultaneously, thereby improving the tightening efficiency:

[0015] In some embodiments, the tightening device further includes a pitch adjustment mechanism, the pitch adjustment mechanism being connected to at least one tightening mechanism and configured to adjust the spacing between at least two tightening mechanisms.

[0016] When tightening operations are performed on assemblies to be tightened provided with different models, due to various dimensions of the assemblies to be tightened, the positions to be tightened of each assembly to be tightened are different. In the case where there are numerous tightening mechanisms, the pitch adjustment mechanism may be configured to adjust the spacing between at least two tightening mechanisms, allowing the tightening mechanisms to adapt to the tightening operations at different positions to be tightened, thereby enhancing the adaptability of the tightening apparatus.

[0017] In some embodiments, the tightening mechanism includes a mechanism body and a lifting mechanism, where the lifting mechanism is connected to the mechanism body and is configured to drive the mechanism body to perform a lifting motion.

[0018] By the arrangement of the lifting mechanism to drive the mechanism body to perform a lifting motion, in an aspect, the mechanism body is capable of descending to a position proximal to the assembly to be tightened to facilitate the tightening operation, and in another aspect, the mechanism body is capable of being lifted up to reduce the damage caused by collision between the mechanism body and other devices when the moving device drives the tightening device to move.

[0019] In some embodiments, the moving device is capable of driving the vision device to move along a preset direction to identify a plurality of positions to be tightened of the assembly to be tightened on the loading station.

[0020] The moving device is configured to control the tightening device to sequentially move right above the plurality of positions to be tightened according to the plurality of positions to be tightened of the assembly to be tightened that are identified by the vision device, and the tightening device tightens the respective tightening component to each position to be tightened.

[0021] In the case where there are a plurality of positions to be tightened, the moving device is arranged to drive the vision device to move along a preset direction to acquire the plurality of positions to be tightened of the assembly to be tightened. Subsequently, according to the plurality of positions to be tightened, the moving device controls the tightening device to sequentially move right above the plurality of positions to be tightened, and the tightening device tightens the respective tightening component to each position to be tightened, which can satisfy sequential tightening operations on the plurality of positions to be tightened and increase the tightening success rate, thereby improving the tightening efficiency.

[0022] In some embodiments, the vision device includes at least two cameras, all of the cameras being configured to jointly identify at least two positions to be tightened of the assembly to be tightened on the loading station.

[0023] It is not only simple to arrange at least two cameras to identify the positions to be tightened of the assembly to be tightened on the loading station, but also has a wider field of view provided

by at least two cameras, which is conducive to improving the reliability of the identification of the positions to be tightened and may also adapt to an assembly to be tightened which is provided with a larger dimension.

[0024] In some embodiments, the tightening apparatus further includes a gripping device, the gripping device being capable of being arranged on the moving device:

[0025] the gripping device is configured to grip a first component to be tightened in the assembly to be tightened at a gripping position, and place the first component to be tightened at a placement position on the loading station, such that the first component to be tightened is assembled with other components to be tightened in the assembly to be tightened on the loading station.

[0026] By the arrangement of the gripping device, the first component to be tightened can be automatically gripped and assembled with other components to be tightened, which can improve the assembly precision, thereby increasing the tightening success rate. In addition, the gripping device and the tightening device share the same moving device, which simplifies the overall structure of the tightening apparatus and saves space.

[0027] In some embodiments, the moving device is capable of mating with one of the gripping device or the tightening device.

[0028] The operations of the gripping device and the tightening device cannot be performed simultaneously, therefore, when both the gripping device and the tightening device are arranged on the moving device at the same time, in an aspect, the independent operation of the gripping device and the tightening device is affected, and in another aspect, the load on the moving device is also increased. Therefore, by arranging the moving device to mate with one of the gripping device or the tightening device, the moving device is capable of detaching the tightening device and only mating with the gripping device when the gripping device is required to grip the first component to be tightened: while the moving device is capable of detaching the gripping device and only mating with the tightening device when the tightening device is required for a tightening operation. This not only improves the reliability of the independent operation of the gripping device and the tightening device, but also reduces the load on the moving device.

[0029] In some embodiments, the tightening device includes a first mating structure, and the moving device includes a switching mechanism, the switching mechanism being capable of mating with or detaching from the first mating structure: [0030] the gripping device includes a second mating structure, and the switching mechanism is also capable of mating with or detaching from the second mating structure.

[0031] In this way, the switching mechanism is capable of mating with or detaching from the first mating structure and the second mating structure to improve the reliability and the precision of the moving device's mating with the tightening device and the gripping device.

[0032] In some embodiments, the tightening apparatus further includes a pressing device, the pressing device being arranged on the loading station: [0033] the pressing device is configured to press the assembly to be tightened on the loading station or to detach from the assembly to be tightened.

[0034] By using the pressing device to press the assembly to be tightened on the loading station, the assembly to be tightened can be fixed before the tightening device performs the tightening operation, which reduces the force exerted on the assembly to be tightened during the tightening process and prevents positional shifts between components that could lead to tightening failure. Therefore, the tightening success rate is improved.

[0035] In some embodiments, the pressing device includes pressing assemblies, and each pressing assembly includes pressing blocks, a telescoping mechanism, and a pressing drive mechanism, where the telescoping mechanism is connected to the pressing drive mechanism and the pressing blocks, the telescoping mechanism is configured to drive the pressing blocks to perform a telescoping motion, and the pressing drive mechanism is configured to drive the pressing blocks to perform a pressing motion to press the assembly to be tightened on the loading station, a direction

of the pressing motion intersecting with the direction of the telescoping motion.

[0036] By the arrangement of connecting the pressing drive mechanism to the pressing blocks, the pressing blocks are capable of pressing the assembly to be tightened on the loading station, which simplifies the pressing process. By the arrangement of connecting the telescoping mechanism to the pressing blocks, in an aspect, when the telescoping mechanism extends, the pressing blocks can accurately press the assembly to be tightened, and in another aspect, when the telescoping mechanism retracts, the placement of the first component to be tightened by the aforementioned gripping device at the placement position is not affected, and the identification of the position to be tightened of the assembly to be tightened by a vision device on the loading station is not affected either.

[0037] In some embodiments, there are a plurality of pressing blocks, the clearance space is formed between at least two adjacent pressing blocks, and the clearance space is configured to expose the position to be tightened of the assembly to be tightened.

[0038] By the arrangement of forming the clearance space between at least two adjacent pressing blocks that are spaced apart, the impact of the pressing blocks on the tightening operation by the tightening device on the assembly to be tightened is reduced, which makes the tightening operation more reliable.

[0039] In some embodiments, the pressing device includes at least two sets of pressing assemblies, and the at least two sets of pressing assemblies are oppositely arranged along the direction of the telescoping motion.

[0040] In this way, pressing can be formed on at least two opposite sides of the assembly to be tightened, thereby improving the reliability of the pressing process.

[0041] In some embodiments, the loading station further includes a transport platform and a positioning mechanism, where the transport platform is capable of bearing and transporting the assembly to be tightened, and the positioning mechanism is capable of positioning the assembly to be tightened on the transport platform.

[0042] Typically, the transport platform is capable of directly interfacing with and transporting the assembly to be tightened to the transport mechanism of the tightening apparatus, such that the manufacturing continuity of the product corresponding to the assembly to be tightened is enhanced, thereby improving the production efficiency; while the arrangement of the positioning mechanism ensures that the assembly to be tightened is more reliably positioned on the transport platform to facilitate corresponding operations on the assembly to be tightened by other devices of the tightening apparatus.

[0043] In a second aspect, provided is a battery manufacturing system, which includes the tightening apparatus according to any of the above embodiments.

[0044] The aforementioned battery manufacturing system operates as follows: after the loading station bears the assembly to be tightened, the moving device may drive the vision device proximal to the assembly to be tightened to identify the positions to be tightened of the assembly to be tightened: according to the positions to be tightened, the moving device may control the tightening device to move to the positions to be tightened, and subsequently, the tightening device tightens the tightening components to the positions to be tightened of the assembly to be tightened, thereby completing the tightening operation. Since the positions to be tightened of the assembly to be tightened are identified by the vision device before a tightening operation, the moving device is capable of accurately driving the tightening device to be right above the positions to be tightened to tighten the tightening components to the positions to be tightened of the assembly to be tightened with success. Therefore, the tightening failure rate of the tightening components is reduced, and due to the use of the tightening device that automatically tightens the assembly to be tightened, the tightening efficiency is also improved.

[0045] The above description is only an overview of the technical solution of the present application. To more clearly understand the technical means of the present application to enable

implementation in accordance with the content of the specification and to make the above and other purposes, features, and advantages of the present application more obvious and easy to understand, the detailed description of the present application is provided below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0046] To more clearly illustrate the technical solutions in the embodiments of the present application, the drawings required for illustrating the embodiments of the present application are briefly described below. Apparently, the drawings in the following description illustrate merely some embodiments of the present application, and those of ordinary skills in the art may still derive other drawings from these drawings without creative efforts.

[0047] FIG. 1 is a schematic structural diagram of a tightening apparatus according to one or a plurality of embodiments.

[0048] FIG. 2 is a schematic structural diagram of a gripping device at a placement position within a tightening apparatus in FIG. 1.

[0049] FIG. 3 is a schematic structural diagram of a tightening device located right above positions to be tightened within the tightening apparatus in FIG. 1.

[0050] FIG. 4 is a schematic structural diagram of a tightening device placed on a support within the tightening apparatus in FIG. 1.

[0051] FIG. 5 is a schematic side surface structural diagram of the tightening device placed on the support shown in FIG. 4.

[0052] FIG. 6 is a schematic structural diagram of a gripping device placed on a placement support within the tightening apparatus in FIG. 1.

[0053] FIG. 7 is a schematic structural diagram of a pressing device within the tightening device in FIG. 1.

[0054] FIG. 8 is an enlarged schematic structural diagram of a part A within the pressing device shown in FIG. 7.

REFERENCE NUMERALS IN THE DETAILED DESCRIPTION ARE AS FOLLOWS

[0055] A tightening apparatus **100**, a loading station **10**, a transport platform **11**, a transport frame **111**, a transport wheel **112**, a positioning mechanism **12**, a moving device **20**, a tightening device **30**, a tightening mechanism **31**, a mechanism body **311**, a lifting mechanism **312**, a pitch adjustment mechanism **32**, a translation cylinder **321**, a guide rail **322**, a slider **323**, a tightening support **33**, a base **331**, a vertical frame **332**, a first mating structure **34**, a vision device **40**, a camera **41**, a gripping device **50**, a suction cup **51**, a sub-suction cup **511**, a second mating structure **52**, a placement support **60**, a pressing device **70**, a pressing assembly **71**, a pressing block **711**, a telescoping mechanism **712**, a pressing drive mechanism **713**, a first connecting plate **714**, a second connecting plate **715**, a guide rod **716**, a buffer member **717**, an assembly **200** to be tightened, a first component **210** to be tightened, and a second component **220** to be tightened.

DETAILED DESCRIPTION

[0056] Embodiments of the technical solutions of the present application will be described in detail below with reference to the drawings. The following embodiments are only for illustrating the technical solutions of the present application more clearly, and therefore are only exemplary and do not limit the protection scope of the present application.

[0057] Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by those skilled in the technical field to which the present application belongs. The terms used herein are only for illustrating the specific embodiments, rather than limiting the present application. The terms “include”, “comprise” and “provided with”, and any variations thereof in the specification and claims of the present application and the above-

mentioned drawing description encompass non-exclusive inclusions.

[0058] In the description of the embodiments of the present application, technical terms such as “first”, “second”, and the like are only used to distinguish different objects and should not be interpreted as indicating or implying the relative importance or implicitly indicating the number, specific order, or primary and secondary relationship of the indicated technical features. In the description of the embodiments of the present application, unless otherwise specifically defined, “a plurality of” means two or more than two.

[0059] Reference in the present application to “embodiment” means that a particular feature, structure, or characteristic described in combination with the embodiment can be included in at least one embodiment of the present application. The references of the word in the context of the specification do not necessarily refer to the same embodiment, nor to separate or alternative embodiments exclusive of other embodiments. It will be explicitly and implicitly appreciated by those skilled in the art that the embodiments described herein can be combined with other embodiments.

[0060] In the description of the embodiments of the present application, the term “and/or” is merely a way to describe the associative relationship between associated objects, indicating that there are three possible relationships. For example, “1 and/or 2” may denote: the presence of 1 alone, the simultaneous presence of 1 and 2, and the presence of 2 alone. In addition, the character “/” herein generally indicates an “or” relationship between the associated objects before and after the “/”.

[0061] In the description of the embodiments of the present application, the term “a plurality of” refers to more than two (including two). Similarly, “multiple groups” refers to more than two groups (including two groups), and “multiple pieces” refers to more than two pieces (including two pieces).

[0062] In the description of the embodiments of the present application, the technical terms “center”, “longitudinal”, “transverse”, “length”, “width”, “thickness”, “upper”, “lower”, “front”, “rear”, “left”, “right”, “vertical”, “horizontal”, “top”, “bottom”, “inner”, “outer”, “clockwise” “counterclockwise”, “axial”, “radial”, “circumferential” and the like indicating directional or positional relationships are based on the directional or positional relationships shown in the drawings. They are merely for the convenience of describing the embodiments of the present application and simplifying the description, and are not intended to indicate or imply that the devices or elements referred to must have specific directions, be constructed and operated in specific directions. Therefore, these terms should not be construed as limitations on the embodiments of the present application.

[0063] In the description of the embodiments of the present application, unless otherwise clearly specified and defined, the technical terms “mount”, “interconnect”, “connect”, “fix”, and the like should be interpreted in their broad senses. For example, “connect” may be “fixedly connect”, “detachably connect”, or “integrally connect”: “mechanically connect” or “electrically connect”: or “directly interconnect”, “indirectly interconnect through an intermediate”, “communication between interiors of two elements”, or “interaction between two elements”. For those of ordinary skill in the art, the specific meanings of the above terms in the embodiments of the present application can be interpreted according to the specific condition.

[0064] At present, batteries on the market typically use metallic cases, and a case structure is typically formed by combining a case body located below and a cover located above. In the battery manufacturing process, a battery cell and related electrical elements are typically arranged in the case body first, and then the cover is assembled with the case body and is tightened by screws, ultimately forming the case.

[0065] In the related art, the cover and the case body are typically combined in a manual method, and then a tightening gun is used to tighten the screws onto the combined cover and case body. However, due to the large number of screw positions on both the cover and the case body, manual tightening often relies on visual identification of the screw positions without precise positioning,

and thus the screws are prone to misalign during the tightening process. As a result, the tightening failure rate of the screws is high and when misalignment occurs, the screws need to be removed and re-tightened or replaced, which reduces the tightening efficiency.

[0066] To relieve the problems of high tightening failure rate of the screws and low tightening efficiency in the related art, a tightening apparatus is designed according to the embodiments of the present application. The tightening apparatus includes a loading station, a moving device, a tightening device, and a vision device. The loading station is configured to bear an assembly to be tightened, the moving device is capable of moving relative to the loading station, the tightening device is arranged on the moving device, and the vision device is arranged on the moving device and is configured to identify positions to be tightened of the assembly to be tightened on the loading station, where the moving device is configured to control the tightening device to move right above the positions to be tightened according to the positions to be tightened of the assembly to be tightened that are identified by the vision device, and the tightening device is configured to tighten tightening components to the positions to be tightened of the assembly to be tightened.

[0067] In this way, after the loading station bears the assembly to be tightened, the moving device may drive the vision device proximal to the assembly to be tightened to identify the positions to be tightened of the assembly to be tightened: according to the positions to be tightened, the moving device controls the tightening device to move right above the positions to be tightened, and subsequently, the tightening device tightens the tightening components to the positions to be tightened of the assembly to be tightened, thereby completing the tightening operation. Since the positions to be tightened of the assembly to be tightened are identified by the vision device before the tightening operation, the moving device is capable of accurately driving the tightening device to move right above the positions to be tightened to tighten the tightening components to the positions to be tightened of the assembly to be tightened with success. Therefore, the tightening failure rate of the tightening components is reduced, and due to the use of the tightening device that automatically tightens the assembly to be tightened, the tightening efficiency is also improved.

[0068] FIG. 1 is a schematic structural diagram of a tightening apparatus according to one or a plurality of embodiments. FIG. 2 is a schematic structural diagram of a gripping device at a placement position within a tightening apparatus in FIG. 1. FIG. 3 is a schematic structural diagram of a tightening device located right above positions to be tightened within the tightening apparatus in FIG. 1. Referring to FIGS. 1 to 3, an embodiment of the present application provides a tightening apparatus **100**, which includes a loading station **10**, a moving device **20**, a tightening device **30**, and a vision device **40**. The loading station **10** is configured to bear an assembly **200** to be tightened, the moving device **20** is capable of moving relative to the loading station **10**, the tightening device **30** is arranged on the moving device **20**, and the vision device **40** is arranged on the moving device **20** and is configured to identify the positions AA to be tightened of the assembly **200** to be tightened on the loading station **10**. The moving device **20** is configured to control the tightening device **30** to move right above the positions AA to be tightened according to the positions AA to be tightened of the assembly **200** to be tightened that are identified by the vision device **40**, and the tightening device **30** is configured to tighten tightening components to the positions AA to be tightened of the assembly **200** to be tightened.

[0069] The loading station **10** is a bearing mechanism configured to bear the assembly **200** to be tightened. The structural form of the loading station **10** is not limited to a loading rack, a loading tray, a loading conveyor belt, or the like.

[0070] The moving device **20** refers to a mechanism that is capable of driving a device attached on the moving device to move. The motion form of the moving device **20** is not limited to translation and rotation, and the moving device may also perform a curvilinear motion along a preset direction, or a combination of a plurality of motion forms. The motion device **20** may be a robot arm, a lifting device, a horizontal moving device, or the like.

[0071] The tightening device **30** and the vision device **40** may be fixedly or detachably arranged on

the moving device **20**. In the embodiments of the present application, the vision device **40** is directly arranged on the moving device **20**. In this way, the vision device can follow the motion of the moving device **20** to correctly acquire the position to be tightened. In other embodiments, the vision device **40** may also be arranged on the tightening device **30**. As such, the vision device may be arranged indirectly on the moving device **20** through the tightening device **30**. It should be noted here that the field of view of the vision device **40** for acquiring the position to be tightened should not be affected when the tightening device **30** is assembled onto the moving device **20**.

[0072] The assembly **200** to be tightened according to the embodiments of the present application may be an assembly combined by a case body and a cover, or may be other assemblies to be tightened. The tightening components according to the embodiments of the present application may be screws or other tightening components. The positions AA to be tightened may be the positions of screw holes.

[0073] After the loading station **10** bears the assembly **200** to be tightened, the moving device **20** may drive the vision device **40** proximal to the assembly **200** to be tightened to identify the positions AA to be tightened of the assembly **200** to be tightened: according to the positions AA to be tightened, the moving device **20** may control the tightening device **30** to move to the positions AA to be tightened; and subsequently, the tightening device **30** tightens the tightening components to the positions AA to be tightened of the assembly **200** to be tightened, thereby completing the tightening operation. Since the positions AA to be tightened of the assembly **200** to be tightened are identified by the vision device **40** before the tightening operation, the moving device **20** is capable of accurately driving the tightening device **30** to be right above the positions AA to be tightened to tighten the tightening components to the positions AA to be tightened of the assembly **200** to be tightened with success. Therefore, the tightening failure rate of the tightening components is reduced, and due to the use of the tightening device **30** that automatically tightens the assembly **200** to be tightened, the tightening efficiency is also improved.

[0074] Referring to FIGS. **1** and **2**, according to some embodiments of the present application, the loading station **10** includes a transport platform **11** and a positioning mechanism **12**. The transport platform **11** is capable of bearing and transporting the assembly **200** to be tightened, and the positioning mechanism **12** is capable of positioning the assembly **200** to be tightened on the transport platform **11**.

[0075] The transport platform **11** refers to a mechanism provided with both transportation and bearing functions. After the assembly **200** to be tightened is placed on the transport platform **11**, the transport platform **11** is capable of transporting the assembly **200** to be tightened to a preset position.

[0076] The positioning mechanism **12** is a mechanism that is capable of positioning the assembly **200** to be tightened on the transport platform **11** at a preset position. The positioning mechanism **12** may position the assembly **200** to be tightened by blocking the motion of the assembly **200** to be tightened on the transport platform **11**. Specifically, the positioning mechanism **12** may include a stop block, a stop plate, or the like.

[0077] Typically; the transport platform **11** is capable of directly interfacing with and transporting the assembly **200** to be tightened to the transport mechanism of the tightening apparatus **100**, such that the manufacturing continuity of the product corresponding to the assembly **200** to be tightened is enhanced, thereby improving the production efficiency; while the arrangement of the positioning mechanism **12** ensures that the assembly **200** to be tightened is more reliably positioned on the transport platform **11** to facilitate corresponding operations on the assembly **200** to be tightened by other devices of the tightening apparatus **100**.

[0078] Specifically, the transport platform **11** includes a transport frame **111** and a plurality of transport wheels **112**, each of the plurality of transport wheels **112** at least partially protrudes from the surface of the transport frame **111** to bear and transport the assembly **200** to be tightened.

[0079] The plurality of transport wheels **112** may actively rotate to transport the assembly **200** to be

tightened, or a pushing drive component may be used to push the assembly **200** to be tightened to slide over the plurality of transport wheels **112**.

[0080] In some embodiments, the positioning mechanism **12** includes a telescoping component that is capable of performing a telescoping motion. When the positioning mechanism **12** positions the assembly **200** to be tightened on the transport platform **11**, the telescoping component is capable of extending out of the surface of the transport platform **11** to abut against one side of the assembly **200** to be tightened.

[0081] Specifically, there are at least two telescoping components, which are spaced apart from each other along the transport direction of the transport platform **11**. A positioning location for the assembly **200** to be tightened is formed between at least two telescoping components.

[0082] The telescoping component may be retracted when positioning of the assembly **200** to be tightened is not required.

[0083] In some embodiments, the positioning mechanism **12** may be arranged below the transport platform **11**. The transport platform **11** is provided with a communication hole for the telescoping component to pass through. The telescoping component is capable of extending out of or retracting into the surface of the transport platform **11** through the communication hole.

[0084] Referring to FIG. **1**, according to some embodiments of the present application, the moving device **20** includes a robot arm, and both the tightening device **30** and the vision device **40** are arranged on the robot arm.

[0085] Compared with other moving devices, the robot arm occupies less space, offers greater flexibility in motion, and is more conducive to precisely reaching the position to be tightened, thereby improving the reliability of the tightening process.

[0086] Referring to FIGS. **3** and **4**, according to some embodiments of the present application, the tightening device **30** includes a plurality of tightening mechanisms **31**, all of the tightening mechanisms **31** are spaced apart from each other, and each tightening mechanism **31** is capable of tightening the tightening component to the corresponding position AA to be tightened of the assembly **200** to be tightened.

[0087] By the arrangement of the plurality of tightening mechanisms **31**, a plurality of positions AA to be tightened of the assembly **200** to be tightened can be tightened simultaneously, thereby improving the tightening efficiency:

[0088] It should be noted here that the vision device **40** may identify a plurality of positions AA to be tightened of the assembly **200** to be tightened, such that a plurality of tightening mechanisms **31** may simultaneously tighten the plurality of positions AA to be tightened of the assembly **200** to be tightened.

[0089] In the embodiments of the present application, the tightening device **30** includes two tightening mechanisms **31**, and the two tightening mechanisms **31** are spaced apart from each other.

[0090] Further, the tightening device **30** further includes a pitch adjustment mechanism **32**, the pitch adjustment mechanism **32** being connected to at least one tightening mechanism **31** and configured to adjust the spacing between at least two tightening mechanisms **31**.

[0091] When tightening operations are performed on assemblies **200** to be tightened with different models, due to various dimensions of the assemblies **200** to be tightened, the positions AA to be tightened of each assembly **200** to be tightened are different. In the case where there are numerous tightening mechanisms **31**, the pitch adjustment mechanism **32** may be used to adjust the spacing between at least two tightening mechanisms **31**, allowing the tightening mechanisms **31** to adapt to the tightening operations at different positions AA to be tightened, thereby enhancing the adaptability of the tightening apparatus **100**.

[0092] Specifically, the pitch adjustment mechanism **32** may be a translational pitch adjustment mechanism that is horizontally movable to adjust the spacing between at least two tightening mechanisms **31**.

[0093] In the embodiments of the present application, the pitch adjustment mechanism **32** includes

a translation cylinder **321**, and the translation cylinder **321** is connected to one tightening mechanism **31** and configured to push the tightening mechanism **31** distal to or proximal to another adjacent tightening mechanism **31**.

[0094] The translational motion of the translation cylinder **321** is smooth, such that the pitch adjustment between the tightening mechanism **31** and another tightening mechanism **31** is smooth.

[0095] Further, the pitch adjustment mechanism **32** further includes a guide rail **322** and a slider **323**, where the slider **323** is connected to the translation cylinder **321** and the tightening mechanism **31**, the slider **323** is in sliding fit with the guide rail **322**, and the guide rail **322** is configured to guide the slider **323** to move along the translation direction of the translation cylinder **321**.

[0096] By arrangement of the guide rail **322** and the slider **323**, the stability of the translational motion of the tightening mechanism **31** in the translation cylinder **321** can be improved.

[0097] In some embodiments, the tightening device **30** includes tightening supports **33**, and all of the tightening mechanisms **31** and the pitch adjustment mechanism **32** may be arranged on the tightening supports **33**.

[0098] In order not to affect the tightening operations of the tightening mechanisms **31**, all of the tightening mechanisms **31** may be arranged on the same side of the tightening supports **33**, and the pitch adjustment mechanism **32** may be arranged on one side of the tightening support **33** away from the tightening mechanism **31**.

[0099] Specifically, the tightening support **33** includes a base **331** and a vertical frame **332**, where the vertical frame **332** is located on one side of the base **331**, and the bottom of the vertical frame is connected to the base **331**. All of the tightening mechanisms **31** are arranged outside the vertical frames **332**, and the pitch adjustment mechanism **32** is arranged on the base **331**.

[0100] Referring to FIG. 5, according to some embodiments of the present application, the tightening mechanism **31** includes a mechanism body **311** and a lifting mechanism **312**, where the lifting mechanism **312** is connected to the mechanism body **311** and configured to drive the mechanism body **311** to perform a lifting motion.

[0101] Specifically, the direction of the lifting mechanism **312** driving the mechanism body **311** to perform a lifting motion is the Z direction shown in FIG. 1. The pitch adjustment mechanism **32** adjusts the spacing direction between at least two tightening mechanisms **31** to be perpendicular to the lifting motion direction of the lifting mechanisms **312**.

[0102] By arranging the lifting mechanism **312** to drive the mechanism body **311** to perform a lifting motion, in an aspect, the mechanism body **311** is capable of descending to a position proximal to the assembly **200** to be tightened to facilitate the tightening operation, and in another aspect, the mechanism body **311** is capable of being lifted up to reduce the damage caused by collision between the mechanism body **311** and other devices when the moving device **20** drives the tightening device **30** to move.

[0103] In some embodiments, the lifting mechanism **312** may include a lifting drive cylinder, and in other ways, may also be a lifting drive hydraulic cylinder, or the like.

[0104] In some embodiments, the mechanism body **311** may be a tightening gun, and the tightening gun is capable of tightening the tightening component to the position AA to be tightened of the assembly **200** to be tightened. The tightening gun may be an electric tightening gun or a start-up tightening gun.

[0105] Specifically, when the tightening gun works to drive the tightening component to be tightened to a preset torque, the tightening component may be tightened.

[0106] Referring to FIGS. 1 and 3, according to some embodiments of the present application, the moving device **20** is capable of driving the vision device **40** to move along a preset direction to identify a plurality of positions AA to be tightened of the assembly to be tightened **200** on the loading station **10**. The moving device **20** is configured to control the tightening device **30** to sequentially move right above the plurality of positions AA to be tightened according to the plurality of positions AA to be tightened of the assembly **200** to be tightened that are identified by

the vision device **40**, and the tightening device **30** tightens the respective tightening component to each position AA to be tightened.

[0107] Generally, there are a plurality of positions AA to be tightened of the assembly **200** to be tightened, such that the assemblies **200** to be tightened have a close connection relationship: if all of the positions AA to be tightened are simultaneously tightened by the tightening device **30**, once one of the positions AA to be tightened fails to be tightened, all of the positions AA to be tightened need to be re-tightened, which may cause a reduction in the tightening efficiency. Therefore, in the case where there are a plurality of positions AA to be tightened, the moving device **20** is arranged to drive the vision device **40** to move along a preset direction to acquire the plurality of positions AA to be tightened of the assembly **200** to be tightened. Subsequently, according to the plurality of positions AA to be tightened, the moving device **20** controls the tightening device **30** to sequentially move right above the plurality of positions AA to be tightened, and the tightening device **30** tightens the respective tightening component to each position AA to be tightened, which can satisfy sequential tightening operations on the plurality of positions AA to be tightened and increase the tightening success rate, thereby improving the tightening efficiency.

[0108] For example, when there are seven positions AA to be tightened of the assembly **200** to be tightened that are spaced apart from each other along the length direction of the assembly **200** to be tightened, the moving device **20** is capable of driving the vision device **40** to move along the length direction of the assembly **200** to be tightened to sequentially identify the seven positions AA to be tightened of the assembly **200** to be tightened, the moving device **20** then controls the tightening device **30** to firstly move to the first position AA to be tightened according to the seven positions AA to be tightened, and the tightening device **30** performs the corresponding tightening operation. After the tightening is completed, the moving device **20** drives the tightening device **30** to move to the second position AA to be tightened, and the tightening device **30** performs the corresponding tightening operation, and so on, until the tightening operation on the seventh position AA to be tightened is completed.

[0109] Referring again to FIG. **1**, according to some embodiments of the present application, the vision device **40** includes at least two cameras **41**, all of the cameras **41** being configured to jointly identify at least two positions AA to be tightened of the assembly **200** to be tightened on the loading station **10**.

[0110] The cameras **41** may acquire image information of the assembly **200** to be tightened in a photographing manner, and then identify the positions AA to be tightened of the assembly **200** to be tightened based on the image information.

[0111] It is not only simple to arrange at least two cameras **41** to identify the positions AA to be tightened of the assembly **200** to be tightened on the loading station **10**, but also has a wider field of view provided by at least two cameras **41**, which is conducive to improving the reliability of the identification of the positions to be tightened and may also adapt to an assembly **200** to be tightened which is provided with a larger dimension.

[0112] Specifically, all of the cameras **41** are spaced apart from each other. The arrangement of all the cameras **41** may be determined according to the positions AA to be tightened on the assembly **200** to be tightened. For example, in the embodiments of the present application, there are two cameras **41**, and when the moving device **20** drives the cameras **41** to the photographing position, the two cameras **41** are spaced apart from each other along the width direction of the assembly **200** to be tightened, such that the two cameras **41** are capable of identifying at least two positions AA to be tightened in the width direction of the assembly **200** to be tightened.

[0113] In some embodiments, each camera **41** is configured to acquire one position AA to be tightened of the assembly **200** to be tightened.

[0114] Referring to FIGS. **1**, **2** and **6**, according to some embodiments of the present application, the tightening apparatus **100** further includes a gripping device **50**, the gripping device **50** being capable of being arranged on the moving device **20**. The gripping device **50** is configured to grip a

first component **210** to be tightened in the assembly **200** to be tightened at a gripping position, and place the first component **210** to be tightened at a placement position on the loading station **10**, such that the first component **210** to be tightened is assembled with other components to be tightened in the assembly **200** to be tightened on the loading station.

[0115] It should be understood that the gripping position and the placement position are two different positions, and the first component **210** to be tightened may be placed in the gripping position in advance for the gripping device **50** to grip. For example, the first component **210** to be tightened is a cover body, the gripping position may be a storage position where the first components **210** to be tightened are stacked, and the gripping device **50** may grip the first component **210** to be tightened stacked uppermost at a time, but of course, only one first component **210** to be tightened may be placed in the gripping position.

[0116] Other components to be tightened in the assembly **200** to be tightened may be placed in the placement position in advance, such that the gripping device **50** can assemble the gripped first component **210** to be tightened with other components to be tightened in the placement position, thus facilitating the subsequent tightening operation. For example, the assembly **200** to be tightened further includes a second component **220** to be tightened, the second component **220** to be tightened being a case body:

[0117] By arranging the gripping device **50**, the first component **210** to be tightened can be automatically gripped and assembled with other components to be tightened, which can improve the assembly precision, thereby increasing the tightening success rate. Secondly, the gripping device **50** and the tightening device **30** share the same moving device **20**, which simplifies the overall structure of the tightening apparatus **100** and saves space.

[0118] Specifically, the tightening apparatus **100** may be provided with a placement support **60** at the corresponding gripping position, and the first component **210** to be tightened may be placed on the placement support **60** for the gripping device **50** to grip.

[0119] In some embodiments, the gripping device **50** may include a suction cup **51**, and the first component **210** to be tightened is sucked by the suction cup **51**. Specifically, the suction cup **51** may include a plurality of sub-suction cups **511**, the sub-suction cups **511** being spaced apart from each other.

[0120] In other embodiments, the gripping device **50** may also be a jaw, a magnetic attraction component, or the like.

[0121] Referring to FIG. 1, combined with FIGS. 2 and 4, further, the moving device **20** is capable of mating with one of the gripping device **50** or the tightening device **30**.

[0122] The operations of the gripping device **50** and the tightening device **30** cannot be performed simultaneously; therefore, when both the gripping device **50** and the tightening device **30** are arranged on the moving device **20** at the same time, in an aspect, the independent operation of the gripping device **50** and the tightening device **30** is affected, and in another aspect, the load on the moving device **20** is also increased. Therefore, by arranging the moving device **20** to mate with one of the gripping device **50** or the tightening device **30**, the moving device **20** is capable of detaching the tightening device **30** and only mating with the gripping device **50** when the gripping device **50** is required to grip the first component **210** to be tightened; while the moving device **20** is capable of detaching the gripping device **50** and only mating with the tightening device **30** when the tightening device **30** is required for a tightening operation. This not only improves the reliability of the independent operation of the gripping device **50** and the tightening device **30**, but also reduces the load on the moving device **20**.

[0123] Specifically, the tightening device **30** includes a first mating structure **34**, and the moving device **20** includes a switching mechanism, the switching mechanism being capable of mating with or detaching from the first mating structure **34** of the tightening device **30**.

[0124] Similarly, the gripping device **50** includes a second mating structure **52**, and the switching mechanism is capable of mating with or detaching from the second mating structure **52** of the

moving device **20**.

[0125] In this way, the switching mechanism is capable of mating with or detaching from the first mating structure **24** and the second mating structure to improve the reliability and the precision of the moving device **20**'s mating with the tightening device **30** or the gripping device **50**.

[0126] In some embodiments, the switching mechanism may be a suction mechanism that sucks the first mating structure or the second mating structure by suction to mate with the tightening device **30** or the moving device **20**. In other embodiments, the switching mechanism may also be a magnetic attraction mechanism that sucks the first mating structure or the second mating structure by magnetic force to mate with the tightening device **30** or the moving device **20**.

[0127] Referring to FIG. **3**, according to some embodiments of the present application, the tightening apparatus **100** further includes a pressing device **70**, the pressing device **70** being arranged on the loading station **10**. The pressing device **70** is configured to press the assembly **200** to be tightened on the loading station **10** or to detach from the assembly **200** to be tightened.

[0128] The pressing device **70** is capable of pressing the assembly **200** to be tightened, such that the positional relationship between the components of the assembly **200** to be tightened can be maintained and the position of the assembly **200** to be tightened as a whole is fixed.

[0129] The pressing device **70** may press in the combination direction of the components of the assembly **200** to be tightened. For example, when the components of the assembly **200** to be tightened are combined in the vertical direction, the pressing device **70** may press in the vertical direction, when the components of the assembly **200** to be tightened are combined in the horizontal direction, the pressing device **70** may press in the horizontal direction, and of course, the pressing device **70** may also simultaneously press the assembly **200** to be tightened in a plurality of directions.

[0130] By using the pressing device **70** to press the assembly **200** to be tightened on the loading station **10**, the assembly **200** to be tightened can be fixed before the tightening device **30** performs the tightening operation, which reduces the force exerted on the assembly **200** to be tightened during the tightening process and prevents positional shifts between components that could lead to tightening failure. Therefore, the tightening success rate is improved.

[0131] Specifically, the pressing operation of the pressing device **70** may be performed at any time after the gripping device **50** places the first component **210** to be tightened at the placement position to assemble the first component **210** to be tightened with the other components to be tightened in the assembly **200** to be tightened, and before the tightening device **30** tightens the tightening components to the positions AA to be tightened of the assembly **200** to be tightened. In the embodiments of the present application, after the gripping device **50** places the first component **210** to be tightened at the placement position, the pressing device **70** presses the assembly **200** to be tightened on the loading station **10**.

[0132] Referring to FIGS. **7** and **8**, according to some embodiments of the present application, the pressing device **70** includes pressing assemblies **71**, and each pressing assembly **71** includes pressing blocks **711**, a telescoping mechanism **712**, and a pressing drive mechanism **713**. The telescoping mechanism **712** and the pressing drive mechanism **713** are both connected to the pressing blocks **711**, the telescoping mechanism **712** is configured to drive the pressing blocks **711** to perform a telescoping motion, and the pressing drive mechanism **713** is configured to drive the pressing blocks **711** to perform a pressing motion to press the assembly **200** to be tightened on the loading station **10**, the direction of the pressing motion intersecting with the direction of the telescoping motion.

[0133] Specifically, after the gripping device **50** places the first component **210** to be tightened at the placement position, the telescoping mechanism **712** of the pressing assembly **71** drives the pressing blocks **711** to extend, and the pressing drive mechanism **713** of the pressing assembly **71** drives the pressing blocks **711** to press the assembly **200** to be tightened on the loading station **10**. After the tightening device **30** tightens the tightening components to the positions AA to be

tightened, the pressing drive mechanism **713** of the pressing assembly **71** drives the pressing blocks **711** to loosen the assembly **200** to be tightened on the loading station **10**, and the telescoping mechanism **712** of the pressing assembly **71** drives the pressing blocks **711** to retract.

[0134] The direction of the pressing motion may be the Z direction shown in FIG. **1**, and the direction of the telescoping motion may be the Y direction shown in FIG. **1**.

[0135] By connecting the pressing drive mechanism **713** to the pressing blocks **711**, the pressing blocks **711** are capable of pressing the assembly **200** to be tightened on the loading station **10**, thus simplifying the pressing process. By connecting the telescoping mechanism **712** to the pressing blocks **711**, in an aspect, when the telescoping mechanism extends, the pressing blocks **711** can accurately press the assembly **200** to be tightened, and in another aspect, when the telescoping mechanism retracts, the placement of the first component **210** to be tightened by the aforementioned gripping device **50** at the placement position is not affected, and the identification of the position AA to be tightened of the assembly **200** to be tightened by a vision device **40** on the loading station **10** is not affected either.

[0136] Further, there are a plurality of pressing blocks **711**, the clearance space AA is formed between at least two adjacent pressing blocks **711** that are spaced apart, and the clearance space AA is configured to expose the position AA to be tightened of the assembly **200** to be tightened.

[0137] By forming the clearance space between at least two adjacent pressing blocks **711** that are spaced apart, the impact of the pressing blocks **711** on the tightening operation on the assembly **200** to be tightened by the tightening device **30** is reduced, making the tightening operation more reliable.

[0138] In addition, the respective pressing blocks **711** are capable of pressing different positions of the assembly **200** to be tightened on the loading station **10**. Specifically, the respective pressing blocks **711** are sequentially disposed in the X direction shown in FIG. **1**.

[0139] In some embodiments, two adjacent pressing blocks **711** are spaced apart from each other to form a clearance space. In other embodiments, the partial pressing blocks **711** may not be spaced or may be spaced without the clearance space formed therebetween, and thus, the pressing area of the pressing device **70** to the assembly **200** to be tightened may be increased to improve the reliability of the pressing process.

[0140] In some embodiments, the telescoping mechanism **712** is capable of being connected to the pressing blocks **711** via the pressing drive mechanism **713**. That is, when the telescoping mechanism **712** performs a telescoping motion, the pressing drive mechanism **713** is capable of being driven to move to further drive the pressing blocks **711** to move.

[0141] Specifically, the pressing assembly **71** further includes a first connecting plate **714**, and all of the pressing blocks **711** are connected to one end of the first connecting plate **714**. The first connecting plate **714** is connected to a movable end of the pressing drive mechanism **713**. Further, the pressing assembly **71** further includes a second connecting plate **715**, the pressing drive mechanism **713** is mounted on the second connecting plate **715**, and the second connecting plate **715** is connected to a movable end of the telescoping mechanism **712**.

[0142] Both the pressing drive mechanism **713** and the telescoping mechanism **712** may be drive cylinders, drive oil cylinders, or the like that move linearly.

[0143] In order to improve the smoothness of the pressing motion of all the pressing blocks **711** driven by the pressing drive mechanism **713**, in the embodiments of the present application, the pressing assembly **71** further includes a plurality of guide rods **716**, and the plurality of guide rods **716** penetrate through the second connecting plate **715** and are connected to the first connecting plate **714** and configured to guide the first connecting plate **714** to move along the pressing direction.

[0144] In some embodiments, the pressing assembly **71** further includes buffer members **717**, the buffer members **717** being arranged between the first connecting plate **714** and the second connecting plate **715**.

[0145] By arrangement of the buffer members **717**, the acting force between the first connecting plate **714** and the second connecting plate **715** when the pressing drive mechanism **713** drives all the pressing blocks **711** to loosen the assembly **200** to be tightened can be reduced, thus reducing the mechanical damage.

[0146] The buffer member **717** may be a buffer cylinder, a buffer spring, or the like.

[0147] According to some embodiments of the present application, the pressing device **70** includes at least two sets of pressing assemblies **71**, and the at least two sets of pressing assemblies **71** are oppositely arranged along the direction of the telescoping motion.

[0148] In this way, pressing can be formed on at least two opposite sides of the assembly **200** to be tightened, thereby improving the reliability of the pressing process.

[0149] In order to more clearly understand the working principle of the tightening apparatus according to the embodiments of the present application, the working process of the tightening apparatus is described in detail below:

[0150] The loading station **10** loads the second component **220** to be tightened, and after the loading is completed, the moving device **20** mates with the gripping device **50** to grip the first component **210** to be tightened at the gripping position, and the moving device **20** drives the first component **210** to be tightened to move to a position above the second component **220** to be tightened and assemble with the second component **220** to be tightened.

[0151] At this time, the gripping device **50** is detached from the moving device **20** and the moving device **20** then mates with the tightening device **30**. The moving device **20** drives the vision device **40** on the tightening device **30** to a position right above the component **200** to be tightened, and identifies the position AA to be tightened of the assembly **200** to be tightened. After the identification is completed, the pressing device **70** presses the upper part of the first component **210** to be tightened, such that the assembly **200** to be tightened is pressed on the loading station **10**. After the pressing is completed, the tightening device **30** tightens the tightening components to the positions AA to be tightened. After the tightening operation is completed, the pressing device **70** is detached from the assembly **200** to be tightened.

[0152] In addition, the embodiments of the present application further provide a battery manufacturing system, which includes the tightening apparatus **100** in any of the embodiments described above.

[0153] After the loading station **10** bears the assembly **200** to be tightened, the moving device **20** may drive the vision device **40** proximal to the assembly **200** to be tightened to identify the positions AA to be tightened of the assembly **200** to be tightened: according to the positions AA to be tightened, the moving device **20** may control the tightening device **30** to move to the positions AA to be tightened; and subsequently, the tightening device **30** tightens the tightening components to the positions AA to be tightened of the assembly **200** to be tightened, thereby completing the tightening operation. Since the positions AA to be tightened of the assembly **200** to be tightened are identified by the vision device **40** before the tightening operation, the moving device **20** is capable of accurately driving the tightening device **30** to be right above the positions AA to be tightened to tighten the tightening components to the positions AA to be tightened of the assembly **200** to be tightened with success. Therefore, the tightening failure rate of the tightening components is reduced, and due to the use of the tightening device **30** that automatically tightens the assembly **200** to be tightened, the tightening efficiency is also improved.

[0154] In the embodiments of the present application, the assemblies **200** to be tightened are a case body and a case cover.

[0155] In some embodiments, the battery manufacturing system further includes: a module assembling apparatus, configured to assemble battery modules into the case body; and a transfer apparatus, configured to transfer the case body that is assembled with the battery modules to the tightening apparatus **100**, and the tightening apparatus **100** is configured to tighten the case body and the case cover using screws to form a battery pack.

[0156] Further, the battery manufacturing system further includes a testing apparatus, where the testing apparatus is configured to carry out performance tests on the battery pack, including voltage tests, current tests, insulation tests, and the like.

[0157] Finally, it should be noted that the above embodiments are only used to illustrate the technical solutions of the present application, but not to limit them. Although the present application has been described in detail with reference to the foregoing embodiments, those of ordinary skill in the art should understand that they can still modify the technical solutions recorded in the foregoing embodiments or make equivalent substitutions for some or all of the technical features: however, these modifications or substitutions do not make the essence of the corresponding technical solutions deviate from the spirit and scope of the technical solutions in the embodiments of the present application, and these modifications or substitutions shall all fall within the scope of claims and specification of the present application. In particular, the technical features mentioned in the embodiments may be combined in any manner as long as there are no structural conflicts. The present application is not limited to the specific embodiments disclosed herein, but encompasses all technical solutions falling within the scope of the claims.

Claims

1. A tightening apparatus, comprising: a loading station, configured to bear an assembly to be tightened; a moving device, capable of moving relative to the loading station; a tightening device, arranged on the moving device; and a vision device, arranged on the moving device and configured to identify positions to be tightened of the assembly to be tightened on the loading station; wherein the moving device is configured to control the tightening device to move right above the positions to be tightened according to the positions to be tightened of the assembly to be tightened that are identified by the vision device, and the tightening device is configured to tighten tightening components to the positions to be tightened.
2. The tightening apparatus according to claim 1, wherein the tightening device comprises a plurality of tightening mechanisms, all of the tightening mechanisms are spaced apart from each other, and each tightening mechanism is capable of tightening the tightening component to the corresponding position to be tightened.
3. The tightening apparatus according to claim 2, wherein the tightening device further comprises a pitch adjustment mechanism, the pitch adjustment mechanism being connected to at least one of the tightening mechanisms and configured to adjust a spacing between at least two of the tightening mechanisms.
4. The tightening apparatus according to claim 2, wherein the tightening mechanism comprises a mechanism body and a lifting mechanism, the lifting mechanism being connected to the mechanism body and configured to drive the mechanism body to perform a lifting motion.
5. The tightening apparatus according to claim 1, wherein the moving device is capable of driving the vision device to move along a preset direction to identify a plurality of the positions to be tightened of the assembly to be tightened on the loading station; the moving device is configured to control the tightening device to sequentially move right above the plurality of positions to be tightened according to the plurality of positions to be tightened of the assembly to be tightened that are identified by the vision device, and the tightening device tightens the respective tightening component to each position to be tightened.
6. The tightening apparatus according to claim 1, wherein the vision device comprises at least two cameras, all of the cameras being configured to jointly identify at least two of the positions to be tightened of the assembly to be tightened on the loading station.
7. The tightening apparatus according to claim 1, wherein the tightening apparatus further comprises a gripping device, the gripping device being capable of being arranged on the moving device: the gripping device is configured to grip a first component to be tightened in the assembly

- to be tightened at a gripping position, and place the first component to be tightened at a placement position on the loading station, such that the first component to be tightened is assembled with other components to be tightened in the assembly to be tightened on the loading station.
- 8.** The tightening apparatus according to claim 7, wherein the moving device is capable of mating with one of the gripping device or the tightening device.
- 9.** The tightening apparatus according to claim 7, wherein the tightening device comprises a first mating structure, and the moving device comprises a switching mechanism, the switching mechanism being capable of mating with or detaching from the first mating structure: the gripping device comprises a second mating structure, and the switching mechanism is also capable of mating with or detaching from the second mating structure.
- 10.** The tightening apparatus according to claim 1, wherein the tightening apparatus further comprises a pressing device, the pressing device being arranged on the loading station: the pressing device is configured to press the assembly to be tightened on the loading station or to detach from the assembly to be tightened.
- 11.** The tightening apparatus according to claim 10, wherein the pressing device comprises pressing assemblies, and each pressing assembly comprises pressing blocks, a telescoping mechanism, and a pressing drive mechanism, wherein the telescoping mechanism is connected to the pressing drive mechanism and the pressing blocks, the telescoping mechanism is configured to drive the pressing blocks to perform a telescoping motion, and the pressing drive mechanism is configured to drive the pressing blocks to perform a pressing motion to press the assembly to be tightened on the loading station, a direction of the pressing motion intersecting with a direction of the telescoping motion.
- 12.** The tightening apparatus according to claim 11, wherein there are a plurality of the pressing blocks, a clearance space is formed between at least two adjacent pressing blocks, and the clearance space is configured to expose the position to be tightened of the assembly to be tightened.
- 13.** The tightening apparatus according to claim 11, wherein the pressing device comprises at least two sets of the pressing assemblies, and the at least two sets of pressing assemblies are oppositely arranged along the direction of the telescoping motion.
- 14.** The tightening apparatus according to claim 1, wherein the loading station further comprises a transport platform and a positioning mechanism, wherein the transport platform is capable of bearing and transporting the assembly to be tightened, and the positioning mechanism is capable of positioning the assembly to be tightened on the transport platform.
- 15.** A battery manufacturing system, comprising the tightening apparatus according to claim 1.
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